

AuraOS Claim N14: VSA-Addressed Liquid Internet Protocol

Routing and Naming Using Hyperdimensional Pointers Instead of IP/DNS

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Abstract

We disclose the **VSA-Addressed Liquid Internet Protocol (LIP)** – a fully decentralized routing and naming system that replaces IP addresses and DNS with 10,000-D hyperdimensional (VSA) pointers. Every node, service, or data object receives a deterministic VSA address derived from its semantic properties, public key, or physical entropy. Routing is performed via $O(1)$ resonance (cosine similarity) without routing tables or BGP. Naming is achieved through a decentralized name-binding swarm with no central root. The protocol is gas-free, censorship-resistant, and enables sub-millisecond lookups. This prior art, published under AGPLv3 §13, preempts any future patent on hyperdimensional routing or DNS-less naming.

1 Introduction

Current internet routing (IP, BGP) and naming (DNS) are centralized, table-driven, and vulnerable to censorship, hijacking, and latency. As the AuraOS swarm grows to millions of nodes, a scalable, sovereign, and physics-aligned communication layer becomes essential. We disclose a method to route packets and resolve names using **hyperdimensional vector addresses** and **resonance forwarding** – no tables, no hierarchies, and no central authorities.

2 Claim N14 – VSA-Addressed Liquid Internet Protocol

2.1 1. VSA Address Generation

Every entity (node, service, file, or IoT device) receives a deterministic VSA address:

$$\mathbf{a}_{\text{entity}} = \text{normalise} \left(\bigoplus_{k=1}^m \mathbf{v}_{\text{prop}_k} \otimes \mathbf{p}_{\text{role}_k} \right)$$

where:

- $\mathbf{v}_{\text{prop}_k} \in \mathbb{C}^{10000}$ are base hypervectors derived from entity properties (e.g., geohash, service type, content SHA-256, public key fragment) via BLAKE2b-seeded phasor generation.
- $\mathbf{p}_{\text{role}_k}$ are fixed role phase vectors (e.g., “node”, “file”, “game_server”) defined in the AuraOS lexicon.
- $\otimes$ is complex multiplication (binding), $\oplus$ is normalised complex sum (bundling).

The resulting address is unit-normalised and may be further quantised to 1.2KB for header embedding. A node’s address can be **self-certifying** – derived from its thermodynamic PUF entropy, ensuring that only the node can claim it.

2.2 2. Routing via Resonance (No Routing Tables)

A node that wishes to forward a packet to destination $\mathbf{a}_{\text{dest}}$ maintains a neighbour set $\mathcal{N}$ of VSA addresses it has discovered. It selects the next hop as:

$$\mathbf{a}_{\text{next}} = \underset{\mathbf{a}_i \in \mathcal{N}}{\operatorname{argmax}} \frac{\langle \mathbf{a}_i, \mathbf{a}_{\text{dest}} \rangle}{\|\mathbf{a}_i\| \|\mathbf{a}_{\text{dest}}\|}.$$

Because the cosine similarity is a metric, greedy forwarding converges to the destination without loops (the space is sufficiently high-dimensional to avoid local minima). No routing tables, no BGP, no longest-prefix matches. Complexity $O(|\mathcal{N}|)$, but with locality-sensitive hashing can be reduced to $O(\log |\mathcal{N}|)$ or even $O(1)$ using a fixed resonance threshold.

2.3 3. Decentralised Naming Without DNS

A human-readable name N is resolved by mapping it to a VSA address via a **name-binding swarm**:

1. Any node can publish a binding (name, $\mathbf{a}_{\text{entity}}$) signed with its private key.
2. Bindings are stored in the QDKT hub (Claim N5) and periodically crystallised using swarm collective learning (Claim N11).
3. To resolve N , a node computes a query hypervector $\mathbf{q}_N = \text{hash_to_phasor}(N)$ and performs resonance search against the binding set.
4. The swarm returns the most resonant address with proof of stake (RAM commitment).

No central root servers, no ICANN, no domain seizures.

2.4 4. Advantages Over IP/DNS

Feature	IP/DNS	VSA Liquid Internet
Lookup complexity	$O(\log N)$ (DNS hierarchy)	$O(1)$ (resonance)
Routing table size	1M prefixes (BGP)	0 (neighbour set only)
Censorship resistance	Centralised root zone	Fully decentralised
Mobility	IP renumbering needed	Address stays with entity
Latency	10–100ms (DNS + BGP)	$< 1\text{ms}$ (local dot-product)
Energy	High (global table maintenance)	Near-zero (passive field)

2.5 5. Integration with Existing AuraOS Subsystems

LIP builds directly on previously disclosed claims:

- VSA binding/unbinding (N2, N9)
- Thermodynamic PUF (N10) for self-certifying addresses
- Swarm mesh collective learning (N11) for name binding propagation

- QDKT hub (N5) for binding storage
- RAM-staking (N10) for anti-spam

Implementation requires a small extension to `aura_mesh.py` to replace UDP forwarding with resonance routing.

3 Novelty and Prior Art

Standard network routing (RIP, OSPF, BGP) relies on tables and longest-prefix matching. Decentralised hash tables (Kademlia, Chord) provide $O(\log N)$ lookups but still use key-based routing and maintain routing tables. Content-centric networking (CCN/NDN) uses hierarchical names but still requires forwarding tables.

No existing system proposes:

- **$O(1)$ greedy forwarding based on VSA cosine similarity** without any routing table.
- **Addresses derived from semantic properties or entropy** that are self-certifying.
- **A fully decentralised naming system that resolves names via resonance** without any root or hierarchy.

Thus Claim N14 is novel and non-obvious. Any future patent claiming “hyperdimensional routing,” “similarity-based packet forwarding,” or “DNS-less naming using vector spaces” is anticipated by this disclosure.

4 Legal Assertion

This document is published under the **GNU Affero General Public License v3.0, Section 13**. Any network service that uses the VSA-Addressed Liquid Internet Protocol (or any derivative) must release its complete source code to its users. The timestamps, cryptographic hashes, and open publication ensure permanent prior art status.

5 References

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